

SESSION TWO

How to Write a Research *Paper*

From Idea to Research Question

Hosted by Zulqarnain Ali & Muhammad Ibrahim Qasmi · Saturday, 27 June 2026 · 10:00 PM

SESSION
02
Two

RECAP

Where we left off *last time*

01

What research *really is*.

One clear claim, proven, that adds something new.

02

How to *read a paper* without drowning.

Skim, skip what you don't get, look up only what you need.

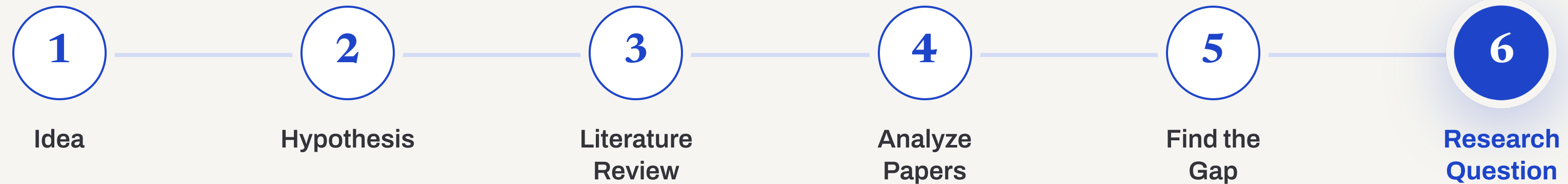


RESEARCH LAB · SLIDES, CODE & RECORDINGS FROM SESSION 1
github.com/muhammadibrahim313/btaji-crew-research-lab

★ Star the repo

WHAT WE WILL DO TODAY

Today we find *your research*.



By the end, you will have *one clear question* of your own.

STEP ONE

Idea Formulation

Research starts with one simple question, something you are curious about. Not a genius plan. **Just a question.**

FOR EXAMPLE

AI models are built for English. So what about Urdu or Punjabi?

STEP TWO

Hypothesis

A hypothesis is your **guess at the answer**, before you check.

FOR EXAMPLE

*Models probably use **more tokens** for our languages than for English.*

STEP THREE

Literature Review

You read what others already did, and find where you fit. Pick papers **close to your question**, and recent ones.

FOR EXAMPLE

🔍 “tokenizer cost languages”

Two close, recent papers come up:

Ahia (2023)

Petrov (2023)

STEP FOUR

For every paper, write *three things*

Problem

What question did they tackle?

Contribution

What did they add or prove?

Limitation

What did they *not* do?

Do this for several papers, not just one.

WORKED EXAMPLE · PAPER ONE

Ahia (2023)

PROBLEM

Are some languages overcharged for tokens?

CONTRIBUTION

Showed yes. Many languages pay more, a Telugu user about $5\times$ an English user.

LIMITATION

One company's pricing, not many tokenizers, and not with the meaning held fixed.

WORKED EXAMPLE · PAPER TWO

Petrov (2023)

PROBLEM

Is the tokenizer itself unfair across languages?

CONTRIBUTION

Showed the unfairness starts **at the tokenizer**, before the model even runs.

LIMITATION

Measured length ratios, not a clean test across many tokenizers with the same meaning.

STEP FIVE

Finding the Gap

A gap is a **hole in what others did**, that you can fill. Look at the limitations you listed, and ask: can I fill one?

THE GAP, IN OUR EXAMPLE

Both papers showed the problem is real, but neither measured it *cleanly across many tokenizers* with the same meaning.

That hole is the gap.

STEP SIX

Research *Question*

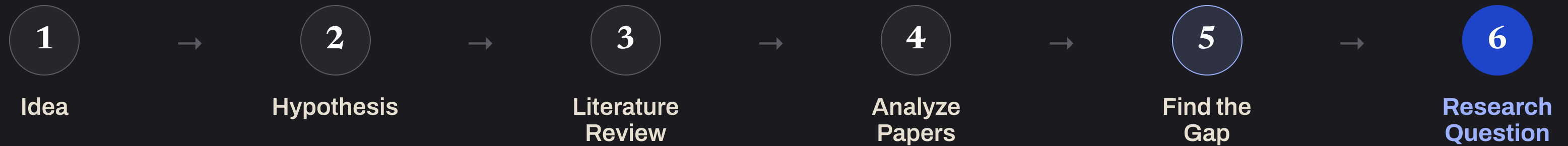
Turn the gap into one clear question you can test. That question is what your **paper answers**.

OUR RESEARCH QUESTION

*How much does token cost change across languages for the **same meaning**, and does it depend on the **tokenizer**?*

THE WHOLE PATH, IN ONE VIEW

From a question, to *your paper*



Each step feeds the next.

The gap becomes the question. The question becomes the paper.

PRACTICAL EXAMPLE

Now Zulqarnain takes over with a *real one*, his own final-year project

FINAL-YEAR RESEARCH PROJECT

Cross-Domain Cross-Modality Transfer Learning via Few-Shot Learning for *Rare Cancer Diagnosis*

LIVE WALKTHROUGH

He will walk through the same six steps, live.

YOUR TASK

Walk the path *yourself*

- 1 Pick a topic you care about, or a recent one.
- 2 Find 10 to 20 recent papers (2022 to 2026).
- 3 For each paper, fill one row in the sheet: **problem, contribution, limitation.**
- 4 Find one gap.
- 5 Write one research question.

FILL A TABLE LIKE THIS

The *format*, on one example

Example topic: using RAG to stop AI from making up facts.

PAPER	PROBLEM	CONTRIBUTION	LIMITATION
Paper 1 (2023)	AI makes up facts	Added a search step to ground answers in real documents	Still fails when the documents are wrong or missing
Paper 2 (2024)	Search brings in the wrong text	A smarter way to pick documents	Slower, and tested only on English

POSSIBLE GAP
Nobody tested this on *our languages*.

These rows just show the format. Use the real papers you find.

JOIN THE COMMUNITY

Keep *going.*

Post your question in the group, and we will help you take the next step.



Join the WhatsApp Community

chat.whatsapp.com/E29f5rozhAo8RbKjA00eSh